INTERNSHIP REPORT

"ExamGuard AI – AI-Powered Online Examination Management Platform"

BACHELOR OF ENGINEERING
in
INFORMATION TECHNOLOGY

By
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(Duration: From / / 2026 to / / 2026)

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ZEAL COLLEGE OF ENGINEERING AND RESEARCH, PUNE

Academic Year 2025-26

ZEAL COLLEGE OF ENGINEERING AND RESEARCH, PUNE

CERTIFICATE

This is to certify that the **Internship Report** submitted by **Om Chandrakant Deo (Roll No.: T51020)** is a record of the work carried out by her during the academic year **2025–2026**, in partial fulfillment of the requirements for the award of the degree of **Bachelor of Engineering in Information Technology**. The internship training was successfully completed at **Rubicon Foundation, Pune**

Internship Coordinator Department	Mr. Aadarsh Shinde **Internship Trainer**	**Head of the Department**

Place: -

Pune

Date: -

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ACKNOWLEDGEMENT

With immense pleasure, I am presenting this report on internship work on **"ExamGuard AI - AI-Powered Online Examination Management Platform"** as a part of the curriculum of T.E. Information Technology at ZEAL COLLEGE OF ENGINEERING AND RESEARCH, PUNE. It gives me proud privilege to complete this report work under the valuable mentorship of Mr. Aadarsh Shinde and Prof. Wrushabh Shirsat.

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Signature

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ABSTRACT

THE COMPANY:

The internship was carried out at **Rubicon Foundation**, a technology-driven organization focused on software product development and digital solutions. The company provided a professional and collaborative environment that enabled hands-on learning in full-stack web development and AI-powered application design, bridging academic concepts with real-world implementation.

PROGRAMMERS AND OPPORTUNITIES:

The internship offered valuable opportunities to work with industry-standard technologies such as **React 18, FastAPI, MongoDB, and cloud platforms**. Collaboration with experienced developers provided practical exposure to modern software engineering practices, including **Agile development, code reviews, version control using Git, and cloud deployment pipelines on Render and Vercel**, thereby enhancing both technical and professional competencies.

METHODOLOGY:

The project was developed using an **Agile iterative methodology**, where tasks were divided into structured weekly sprints. Each sprint focused on delivering specific functional modules—such as authentication, examination management, proctoring, and analytics—followed by thorough testing, integration, and review to ensure system reliability and performance.

KEY PARTS OF THE REPORT:

- Company Background Information
- Introduction to Domain — AI-Powered EdTech & Online Examination Systems
- Responsibilities and Contributions During the Internship
- Experience and Technical Skills Acquired
- Detailed Project Report with Screenshots
- Outcomes and Learnings
- Conclusion and Future Scope

WEEKLY OVERVIEW OF INTERNSHIP ACTIVITIES

	DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED
1st WEEK			

	DATE	DAY	NAME OF THETOPIC/MODULE COMPLETED
2nd WEEK			

3rd WEEK	DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED

4th WEEK	DATE	DAY	NAME OF THE TOPIC/MODULE COMPLETED

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Learning Objectives/Internship Objectives

Company Background Information

Rubicon Foundation is a technology-driven organization based in Delhi, focused on developing innovative software solutions for the education and enterprise sectors. The company specializes in **full-stack web application development, cloud computing, and AI integration**, delivering scalable and efficient digital solutions to a diverse range of clients.

Founded in 2006, the organization has grown into a team of skilled engineers and designers dedicated to building **secure, reliable, and user-friendly applications**. Rubicon emphasizes modern development practices, leveraging emerging technologies to address real-world challenges across industries.

The internship program at Rubicon Foundation is structured to provide engineering students with **hands-on project experience**, enabling them to work on real-world applications under professional mentorship. This approach helps bridge the gap between academic learning and industry expectations while fostering technical and problem-solving skills.

Introduction to domain

The domain of this internship lies in **AI-Powered EdTech**, with a primary focus on advanced **Online Examination Management Systems** and **Learning Management Systems (LMS)**. In recent years, the rapid expansion of digital education, remote learning environments, and global accessibility to online resources has significantly transformed the traditional education landscape. This transformation has created a growing demand for intelligent platforms that not only deliver educational content but also ensure **secure, scalable, and efficient assessment mechanisms**.

Modern EdTech systems are evolving beyond basic content delivery by integrating **Artificial Intelligence (AI)** to enhance both learning and evaluation processes. These platforms now incorporate features such as **automated proctoring, behavioral analytics, adaptive assessments, and real-time performance tracking**, enabling institutions to maintain academic integrity while providing a personalized learning experience.

A critical aspect of this domain is the development of **secure online examination environments**, where technologies like **behavioral biometrics, anomaly detection, and activity monitoring** are used to identify suspicious patterns and prevent malpractice without relying solely on intrusive methods such as video surveillance. This ensures a balance between **privacy, fairness, and security**.

Additionally, Learning Management Systems are becoming more interactive and learner-centric by offering **structured courses, integrated quizzes, coding assessments, progress tracking, and analytics dashboards**. These features help educators make data-driven decisions while allowing students to monitor their growth and improve their skills effectively.

Overall, AI-powered EdTech platforms aim to create a seamless ecosystem that combines **learning, assessment, monitoring, and analytics** into a unified system, thereby enhancing the quality, accessibility, and credibility of digital education.

Responsibilities during the internship

As an intern working on the **ExamGuard AI platform**, I contributed to the development of a full-stack AI-powered examination and learning management system, focusing on backend development, frontend implementation, system integration, and deployment.

I designed and implemented **RESTful API endpoints using FastAPI** to handle core functionalities such as user authentication, exam management, course operations, and submission processing. On the data layer, I developed **MongoDB models using Beanie ODM and Motor**, enabling efficient and asynchronous database interactions for collections like users, exams, courses, and submissions.

I implemented a secure authentication system using **JWT-based role-based access control** for both student and admin portals, along with **bcrypt password hashing** to ensure secure credential management. Additionally, I integrated the **Judge0 API** to support coding assessments, allowing secure execution and evaluation of user-submitted code against predefined test cases.

A key contribution was in building the **proctoring module**, which included tab-switch detection, paste monitoring, and behavioral logging to identify potential malpractice during exams while maintaining a privacy-friendly approach.

On the frontend, I developed responsive and user-friendly interfaces using **React 18 and Tailwind CSS**, covering dashboards, exam pages, and course modules for both admin and student portals. I also implemented **analytics dashboards using Recharts** to visualize performance metrics and system insights.

Furthermore, I participated in **Agile development practices**, including sprint discussions and code reviews, and managed deployment configurations using **Render (backend)** and **Vercel (frontend)** to ensure smooth production readiness.

Through these responsibilities, I gained practical experience in building scalable, secure, and intelligent EdTech solutions across the full development lifecycle.

Experience and skills acquired during internship

Technical Skills Acquired:

- **Full-Stack Development:** Built end-to-end web applications using **React 18 and FastAPI**, ensuring smooth frontend-backend integration.
- **Asynchronous Programming:** Gained practical experience with **async/await** in FastAPI and Motor for efficient, non-blocking operations.
- **Database Design:** Worked with **MongoDB**, designing schemas and implementing models using **Beanie ODM and Pydantic**.
- **Security Implementation:** Implemented **JWT-based authentication**, **bcrypt hashing**, and **role-based access control (RBAC)**.
- **API Integration:** Integrated **Judge0 API** for secure code execution and automated evaluation of coding problems.
- **Cloud Deployment:** Deployed applications on **Render and Vercel**, managing environment configurations and production setup.

Professional Skills Acquired:

- **Agile Development:** Participated in sprint-based workflows, improving adaptability and planning.
- **Version Control:** Used **Git and GitHub** for collaboration, feature branching, and code reviews.
- **Documentation:** Created clear technical documentation and API descriptions.
- **Problem-Solving:** Developed strong debugging and analytical skills while handling real-world issues.
- **Team Collaboration:** Improved communication and teamwork in a professional development environment.

Internship Report & Screenshots

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1. Executive Summary

ExamGuard AI is a full-stack, enterprise-grade online examination platform built during the internship period. The platform provides a complete end-to-end solution for administering, proctoring, and analyzing academic assessments. It combines real-time behavioral analysis, AI-powered test generation, remote code execution, course management, and a secure multi-role authentication system into one unified application.

The system is designed to serve two distinct user roles — **Students** and **Administrators** — each with dedicated portals, protected API layers, and isolated JWT-based authentication

schemes. The backend exposes over **40 REST API endpoints** across 16 route modules. The frontend comprises over **29 React page components** organized across a clean domain-driven architecture.

The project was successfully deployed to production on **Vercel** (frontend) and **Render** (backend), with MongoDB Atlas serving as the cloud database. A comprehensive 10-phase architectural refactoring was also executed to bring the codebase to enterprise production standards.

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2. Introduction

The rise of remote learning and online assessments has exposed critical weaknesses in traditional examination systems: lack of proctoring integrity, manual question paper creation, no real-time monitoring, and limited performance analytics. **ExamGuard AI** was conceived and built to address each of these problems through technology.

The platform is not merely an "online quiz tool." It is a comprehensive academic management system featuring:

- **Behavioral Biometrics Tracking** — Keystroke patterns, paste events, tab switches, and mouse activity are silently logged during exams to generate an **Integrity Risk Score (0–100)** for each student.
- **Live Admin Monitoring** — Administrators can watch student behavior in real-time during examination windows.
- **AI-Powered Test Generation** — Administrators can generate entire exams (MCQ + coding sections) automatically from a curated question bank spanning multiple topics and difficulty levels.
- **Integrated Code IDE** — Students can write, run, and test code directly inside the exam interface, powered by the **Judge0** remote code execution engine.
- **Course Learning Hub** — A complete learning management system (LMS) with lessons, quizzes, and enrollment workflows.

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3. Project Objectives

The following objectives were defined at the start of the internship and successfully fulfilled:

#	Objective	Status
1	Build a secure multi-role authentication system (Student + Admin)	Complete

#	Objective	Status
2	Implement real-time behavioral proctoring with anomaly scoring	Complete
3	Build an AI-powered exam generator from curated question banks	Complete
4	Integrate Judge0 for in-browser code execution	Complete
5	Build a full Course Management / LMS module	Complete
6	Deploy the entire stack to cloud (Vercel + Render + Atlas)	Complete
7	Perform a 10-phase codebase architectural refactoring	Complete
8	Implement Google OAuth2 for Single Sign-On	Complete
9	Build Admin Analytics dashboard with performance metrics	Complete
10	Create comprehensive developer documentation	Complete

---

4. Technology Stack & Justification

4.1 Frontend

Technology	Version	Purpose
React	19.x	Core UI framework with Concurrent mode
Vite	7.x	Build tool — faster than CRA, native ES modules
Tailwind CSS	4.x	Utility-first styling, rapid responsive design
React Router DOM	7.x	Client-side SPA routing with nested layouts
Axios	1.x	HTTP client with interceptors and retry logic
Recharts	3.x	Charting library for analytics dashboards
Lucide React	0.5x	Consistent icon set

Justification: React 19 was chosen for its mature ecosystem and concurrent rendering capabilities. Vite significantly reduces development cycle time compared to Webpack-based

alternatives. Tailwind CSS allows rapid UI prototyping while maintaining strict design consistency.

4.2 Backend

Technology	Version	Purpose
FastAPI	0.135	Async Python REST framework with auto Swagger docs
Python	3.12	Core language — latest stable release
Motor	3.7	Async MongoDB driver for Python
Beanie	2.0	Async ODM/ORM layer over Motor (type-safe models)
Pydantic	2.x	Data validation and serialization
Uvicorn	0.41	ASGI server for production and development
python-jose	3.5	JWT token creation and validation
bcrypt / passlib	—	Secure password hashing
httpx	0.28	Async HTTP client for Google OAuth and Judge0
aiosmtplib	3.0	Async email dispatch

Justification: FastAPI was selected for its first-class `async/await` support, automatic Pydantic validation, and built-in Swagger UI interface for API documentation. Beanie eliminates raw MongoDB query boilerplate while providing full type safety.

4.3 Database

Technology	Purpose
MongoDB Atlas	Cloud NoSQL document database
Motor / Beanie	Async ODM for type-safe queries

Justification: MongoDB's document model naturally fits the deeply nested exam data structures (sections → questions → options → test cases). Atlas provides a globally distributed free-tier cluster ideal for a production deployment.

4.4 Third-Party Integrations

Service	Purpose
Judge0 API	Remote code compilation and test execution
Resend API	Transactional email (password reset, verification)

Service	Purpose
Google OAuth2	Single Sign-On for Student portal
Render	Backend PaaS deployment (Docker-based)
Vercel	Frontend deployment with automatic CI/CD

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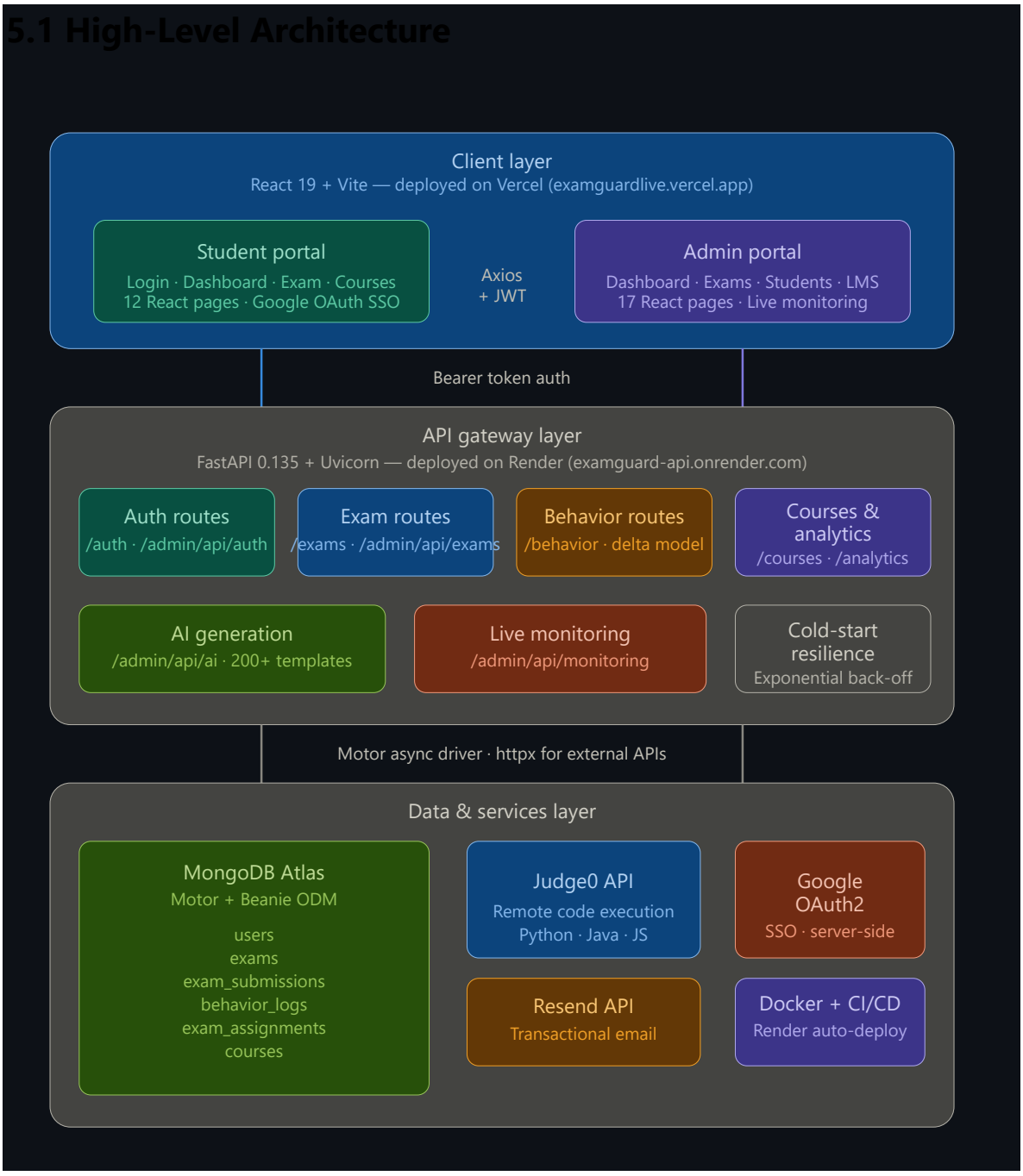
5. System Architecture

5.1 High-Level Architecture



5.2 Frontend Directory Structure

```
frontend/src/
├── admin/
│   ├── context/          # AdminAuthContext.jsx (Admin JWT state)
│   └── pages/            # 17 Admin pages (Dashboard, Exams, Students...)
├── student/
│   ├── context/          # StudentAuthContext.jsx (Student JWT state)
│   └── pages/            # 12 Student pages (Dashboard, ExamPage...)
├── components/
│   ├── admin/            # Sidebar.jsx (Admin navigation)
│   ├── student/          # exam/ sub-components (CodeEditor,
QuestionPalette...)
│   ├── Modal.jsx
│   └── ServerWakingOverlay.jsx
├── hooks/                # 6 Custom React Hooks (useExamTimer,
useProctoringLogger...)
├── services/             # API Singletons (apiClient.js, api.js, adminApi.js)
└── App.jsx               # Root Router + Protected Route wrappers
```

5.3 Backend Directory Structure

```
backend/app/
├── core/
│   ├── database.py        # Motor + Beanie initialization
│   ├── security.py        # JWT creation, password hashing (Student)
│   ├── admin_security.py  # JWT creation (Admin — isolated keys)
│   ├── logging_config.py  # Structured logging setup
│   └── error_handlers.py  # Global 422/500 FastAPI handlers
├── models/
│   ├── all_models.py      # User, Exam, ExamSubmission, BehaviorLog,
ExamAssignment
│   ├── admin_models.py    # Admin account document
│   └── course_models.py   # Course, Lesson, Quiz, Enrollment documents
├── routes/                # 16 route modules (see Section 8)
├── services/
│   ├── email_service.py   # Resend API integration
│   └── ai_test_generation/
│       ├── template_repository.py  # 200+ question bank
│       └── builder.py             # Exam assembly engine
└── utils/
    └── datetime_utils.py  # IST timezone helpers
```

---

6. Database Design

The application uses **MongoDB** with the **Beanie** ODM. Five core document collections are defined:

6.1 User Collection (`users`)

```
User {
  _id           : ObjectId
  username      : String (unique, indexed)
```

```
  email         : String (unique, indexed)
  password_hash : String
  full_name     : String?
  phone_number  : String?
  course        : String?
  college       : String?
  is_active     : Boolean
  created_at    : DateTime (IST)
  auth_provider : "local" | "google"
  google_id     : String?
  reset_token   : String?
  reset_token_expiry : DateTime?
}
```

6.2 Exam Collection (`exams`)

```
Exam {
  _id              : ObjectId
  title            : String
  description      : String?
  sections         : [Section]
  total_marks      : Integer
  duration_minutes : Integer
  start_time       : DateTime (timezone-aware)
  created_at       : DateTime (IST)
}

Section {
  title     : String
  questions : [MCQQuestion | CodingQuestion]
}

MCQQuestion {
  id      : String
  type    : "mcq"
  text    : String
  points  : Integer
  options : [MCQOption]
}

CodingQuestion {
  id                : String
  type              : "coding"
  text              : String
  points            : Integer
  problem_statement : String
  constraints       : String
  test_cases        : [TestCase]
}
```

6.3 Exam Submission Collection (`exam_submissions`)

```
ExamSubmission {
  _id            : ObjectId
  user_id        : String (indexed)
  exam_id        : String (indexed)
  exam_title     : String
  answers        : {section_index: {question_index: answer}}
  status         : "IN_PROGRESS" | "COMPLETED" | "GRADED" | "TERMINATED"
  attempt_number : Integer
  score          : Float?
  mcq_score      : Float?
```

```
  coding_score   : Float?
  anomaly_score  : Integer (0-100)
  risk_level     : "LOW" | "MEDIUM" | "HIGH"
  risk_factors   : [String]
  submitted_at   : DateTime
}
```

6.4 Behavior Log Collection (`behavior_logs`)

```
BehaviorLog {
  _id               : ObjectId
  submission_id     : String
  user_id           : String (indexed)
  exam_id           : String (indexed)
  keystroke_count   : Integer
  avg_typing_speed  : Float (keys/second)
  backspace_ratio   : Float
  paste_count       : Integer
  pasted_chars      : Integer
  tab_switch_count  : Integer
  mouse_click_count : Integer
  time_per_question : {sIdx-qIdx: milliseconds}
  events            : [{type, timestamp}] (max 500)
  recorded_at       : DateTime
}
```

6.5 Exam Assignment Collection (`exam_assignments`)

```
ExamAssignment {
  _id               : ObjectId
  exam_id           : String (indexed)
  assigned_students : [user_id_strings]
  max_attempts      : Integer
  start_time        : DateTime?
  end_time          : DateTime?
}
```

---

7. Key Modules & Features

7.1 Student Portal

The Student Portal provides the complete examination experience for end-users.

Pages:

Page	File	Description
Login	`LoginPage.jsx`	Email/password + Google OAuth SSO
Register	`RegisterPage.jsx`	Account creation with validation
Google Callback	`GoogleCallbackPage.jsx`	OAuth token exchange handler
Forgot Password	`ForgotPasswordPage.jsx`	Email reset request via Resend API

Page	File	Description
Reset Password	`ResetPasswordPage.jsx`	Token-validated password update
Dashboard	`DashboardPage.jsx`	Overview: progress, exams, performance
Waiting Room	`WaitingRoomPage.jsx`	Pre-exam countdown and instructions
Exam	`ExamPage.jsx`	Full exam interface (MCQ + Code IDE)
Test Completed	`TestCompletedPage.jsx`	Score reveal and submission summary
Course List	`CourseList.jsx`	Browse and enroll in courses
Course View	`CourseView.jsx`	Lessons, quizzes, progress tracking
Student Profile	`StudentProfile.jsx`	Profile management and history

Core Student Hooks (in `src/hooks/`):

Hook	Responsibility
`useExamTimer.js`	Countdown timer with auto-submit on expiry
`useExamNavigation.js`	Section/question traversal state management
`useQuestionStatus.js`	Tracks answered/flagged/visited question states
`useProctoringLogger.js`	Captures behavioral events and flushes to backend every 15 seconds
`useExamSubmission.js`	Handles the submit workflow, scoring, and status transitions
`useCodeExecution.js`	Sends code to Judge0 proxy and handles test results

7.2 Admin Portal

The Admin Portal gives instructors full control over the examination ecosystem.

Pages:

Page	File	Description
Login	`Login.jsx`	Admin-specific login with isolated JWT
Register	`Register.jsx`	Create new admin accounts
Dashboard	`Dashboard.jsx`	System-wide KPI overview
Exam Management	`ExamManagement.jsx`	List, search, filter all exams
Create Exam	`CreateExam.jsx`	Manually build exams with sections

Page	File	Description
Edit Exam	`EditExam.jsx`	Modify existing exam content
Preview Exam	`PreviewExam.jsx`	Student-view preview before publishing
Exam Results	`ExamResults.jsx`	Per-exam score distribution + anomaly flags
AI Test Generator	`AITestGenerator.jsx`	Automated exam creation from question bank
Students	`Students.jsx`	Student roster management
Student Progress	`StudentProgress.jsx`	Individual student learning analytics
Course Management	`CourseManagement.jsx`	Create and manage course content
Edit Course	`EditCourse.jsx`	Modify lessons, quizzes, materials
Course Requests	`CourseRequests.jsx`	Approve/reject enrollment requests
Live Monitoring	`LiveMonitoring.jsx`	Real-time proctoring dashboard
Analytics	`Analytics.jsx`	Platform-wide performance analytics
Create Admin	`CreateAdmin.jsx`	Provision new admin accounts

7.3 AI Test Generator

One of the most technically significant features is the AI-powered exam generation system.

How it works:

1. The Admin opens the `AITestGenerator` page and selects:
   - **Topic** (Python, Java, JavaScript, DSA, General CS)
   - **Difficulty Level** (Easy / Medium / Hard)
   - **Number of Aptitude MCQs**, Technical MCQs, and Coding Questions
2. The request is dispatched to `POST /admin/api/ai/generate`
3. The backend's `template_repository.py` module selects questions from over **200+ curated templates** spanning:
   - **Aptitude MCQs:** Speed-Distance, Ratios, Probability, CI/SI, Work-Rate
   - **Technical MCQs:** Python, Java, JavaScript, DSA, General CS
   - **Coding Problems:** Two Sum, Valid Parentheses, Fibonacci, LCS, Merge Intervals, N-Queens
4. The assembled exam JSON is returned to the frontend and pre-fills the Create Exam form

Question Bank Scale:

- **Aptitude Templates:** 22 questions across 3 difficulty tiers

- **Technical Templates:** 55+ MCQs across 5 topics × 3 difficulties
- **Coding Templates:** 25+ problems across Python, Java, and General categories

---

7.4 Behavioral Proctoring Engine

The proctoring system is one of the core differentiators of ExamGuard AI.

Data Captured Per Exam Session:

Signal	Description
`keystroke_count`	Total keystrokes during exam
`avg_typing_speed`	Keys per second (weighted average)
`backspace_ratio`	Backspace : total keystroke ratio
`paste_count`	Number of paste events (Ctrl+V)
`pasted_chars`	Characters introduced via paste
`tab_switch_count`	Number of times student left the exam tab
`mouse_click_count`	Total mouse clicks
`time_per_question`	Milliseconds spent on each question
`events[]`	Raw timeline (up to 500 entries)

Data Flow:

- The `useProctoringLogger` React hook captures events in the browser continuously.
- Every **15 seconds**, a delta payload (only NEW events since last flush) is sent to `POST /behavior/log`.
- The backend **accumulates** deltas into a single `BehaviorLog` document per session.
- On exam submission, the behavior data is analyzed to compute an **Integrity Risk Score (0–100)**.
- Submissions are flagged as `LOW`, `MEDIUM`, or `HIGH` risk with human-readable `risk_factors` and a `risk_explanation`.

7.5 Code Execution Engine

The Exam Page includes a full **Code Editor IDE** for programming-type questions.

Architecture:

```
Student types code
       ↓
useCodeExecution hook sends code to:
POST /exams/execute
       ↓
FastAPI backend proxies to Judge0 API
(https://ce.judge0.com)
       ↓
Judge0 compiles and executes code
Runs against test cases
       ↓
Returns: { passed, actual, expected, status, error }
       ↓
UI renders pass/fail per test case
```

Supported Languages: Any language supported by Judge0, with Language ID passed from the frontend. Default setups include Python 3 (ID: 71), Java (ID: 62), JavaScript (ID: 63).

7.6 Course Management System

The platform includes a complete **Learning Management System (LMS)**:

Student Side:

- Browse available courses with descriptions, lesson counts, and prerequisites
- Submit enrollment requests (pending admin approval)
- Access enrolled courses with structured lessons and video-links
- Track completion with per-lesson checkmarks
- Take module quizzes and see scores

Admin Side:

- Create courses with rich content (markdown descriptions, lesson outlines)
- Manage course modules and their ordering
- View and approve/reject enrollment requests
- Monitor student progress per course through the Analytics dashboard

---

8. API Architecture

The backend exposes **40+ REST endpoints** organized across 16 route module files.

8.1 Route Modules

Module File	Prefix	Description
auth_routes.py	/auth	Student registration, login, OAuth, password reset
exam_routes.py	/exams	Student exam access, start, submit, execute code
behavior_routes.py	/behavior	Proctoring data ingestion (delta model)
course_routes.py	/courses	Course listing, enrollment, progress
course_progress_routes.py	/courses	Lesson completion, quiz submission
student_profile_routes.py	/student/profile	Profile CRUD
admin_auth.py	/admin/api/auth	Admin login, register, me
exam_mgmt.py	/admin/api/exams	Full exam CRUD + assignment
admin_course.py	/admin/api/courses	Course CRUD
admin_enrollment_routes.py	/admin/api	Enrollment approval workflow
student_mgmt.py	/admin/api/students	Student roster management
analytics.py	/admin/api/analytics	Dashboard analytics, exam results
monitoring_routes.py	/admin/api/monitoring	Live proctoring session feeds
admin_progress.py	/admin/api/progress	Student learning progress metrics
ai_generation_routes.py	/admin/api/ai	AI exam generation trigger

8.2 Authentication Design

The system uses **two completely separate JWT authentication schemes** to prevent privilege escalation:

Attribute	Student JWT	Admin JWT
Secret Key	SECRET_KEY env var	ADMIN_SECRET_KEY env var
Algorithm	HS256	HS256
Expiry	1440 minutes (24h)	1440 minutes (24h)
Guard Function	get_current_user()	get_current_admin()

Attribute	Student JWT	Admin JWT
Header	`Authorization: Bearer <token>`	`Authorization: Bearer <token>`

8.3 Cold-Start Resilience

The Render free tier hibernates services after 15 minutes of inactivity.
The `apiClient.js` Axios factory includes:

- Automatic detection of 5xx cold-start errors
- Loading overlay (`ServerWakingOverlay`) shown to users during server wake-up
- Exponential back-off retry logic for failed requests

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9. Security Implementation

Feature	Implementation
Password Hashing	`bcrypt` via `passlib` library
Student JWT	HS256 signed, 24-hour expiry, verified server-side per request
Admin JWT	Separate HS256 key, completely isolated from student tokens
CORS	Strict allowlist via `CORS_ORIGINS` environment variable
Google OAuth	`httpx` async client — token exchanged server-side, user is matched by `google_id`
Password Reset	Time-limited token (stored hash in DB), compared with `secrets.compare_digest`
Timezone Safety	All datetime comparisons use timezone-aware objects (UTC/IST) to prevent `TypeError` crashes
Input Validation	Pydantic V2 schemas validate all request payloads before reaching route handlers
Global Error Handlers	422 Unprocessable Entity and 500 Internal Server Error are caught by `error_handlers.py`

---

10. Deployment & DevOps

10.1 Cloud Infrastructure

Layer	Service	Configuration
Database	MongoDB Atlas (Free M0)	Network access: allow all IPs (for Render)
Backend	Render (Docker)	Root dir: `backend/`, Dockerfile path auto-detected
Frontend	Vercel	Framework: Vite, Root dir: `frontend/`

10.2 Backend Dockerfile Strategy

The backend is containerized with Docker. Render pulls the image directly from the GitHub repository's Dockerfile. Key environment variables are injected via Render's secret management console:

```
MONGODB_URL        = (Atlas connection string)
SECRET_KEY         = (random 256-bit hex string)
ADMIN_SECRET_KEY   = (separate random 256-bit hex string)
CORS_ORIGINS       = https://examguardlive.vercel.app
RESEND_API_KEY     = (Resend dashboard API key)
FRONTEND_URL       = https://examguardlive.vercel.app
GOOGLE_CLIENT_ID   = (Google Cloud Console)
GOOGLE_CLIENT_SECRET = (Google Cloud Console)
GOOGLE_REDIRECT_URI  = https://examguardlive.vercel.app/auth/google/callback
JUDGE0_API_URL     = https://ce.judge0.com
```

10.3 Frontend Environment (Vercel)

```
VITE_USER_API_URL  = https://examguard-api.onrender.com
VITE_ADMIN_API_URL = https://examguard-api.onrender.com/admin/api
```

10.4 CI/CD Pipeline

Both platforms provide automatic CI/CD:

- **Vercel:** Any push to `main` branch triggers a new frontend build and deployment.
- **Render:** Any push to `main` branch triggers a new Docker image build and deployment.

---

11. Codebase Refactoring & Architectural Audit

A comprehensive **10-phase architectural refactoring** was executed as a dedicated workstream to bring the codebase to production enterprise standards.

Phases Completed

Phase	Description	Outcome
Phase 1	Full Codebase Analysis & Mapping	All dead code, orphans, and inconsistencies identified
Phase 2	Remove Unnecessary Files	Deleted 7+ backend diagnostic scripts, orphaned React components
Phase 3	Architectural Restructuring	Flattened nested route dirs; unified services, hooks, components
Phase 4	Code Quality & Consolidation	PascalCase/snake_case enforcement; deduplicated API handlers
Phase 5	Fix Runtime Errors	Patched timezone `TypeError` crashes; hardened API payloads
Phase 6	Configuration Cleanup	Generated `.env.example` for both stacks; removed all hardcoded URLs
Phase 7	Documentation	Created `DEVELOPER_GUIDE.md`, `API_REFERENCE.md`, `DEPLOYMENT_GUIDE.md`
Phase 8	System Testing	Verified Auth flows, Exam rendering, Admin panel operations
Phase 9	Performance	Minimized React re-renders; optimized MongoDB query indexes
Phase 10	Final Validation	`npm run build` completes — 2,500+ modules transformed in 2.53s

Key Structural Changes

Frontend (before → after):

- `src/lib/apiClient.js` → `src/services/apiClient.js`
- `src/admin/components/Sidebar.jsx` → `src/components/admin/Sidebar.jsx`
- `src/admin/services/adminApi.js` → `src/services/adminApi.js`
- `src/student/services/api.js` → `src/services/api.js`
- `src/student/hooks/use*.js` (6 files) → `src/hooks/use*.js`

Backend (before → after):

- `backend/app/db/session.py` → `backend/app/core/database.py`
- `backend/app/routes/routes/` (recursive) → `backend/app/routes/` (flat)
- Production API sanitized of all diagnostic endpoints

---

12. Challenges Faced & Solutions

Challenge 1: Naive Datetime `TypeError` Crash

Problem: MongoDB returns timezone-naive `datetime` objects. Comparing these with timezone-aware Python `datetime.now(UTC)` caused a `TypeError: can't compare offset-naive and offset-aware datetimes` → HTTP 500 crash on the password reset endpoint.
Solution: Implemented a timezone coercion patch in `auth_routes.py`. All datetimes retrieved from MongoDB are explicitly coerced to UTC-aware using `.replace(tzinfo=timezone.utc)` before any comparison operation.

Challenge 2: SMTP Email Failure on Render

Problem: The initial email service used `aiosmtplib` connecting to `smtp.gmail.com:587`. Render's free-tier infrastructure **blocks outbound SMTP on port 587**, causing `TimeoutError`.
Solution: Migrated the email service entirely to the **Resend API** (HTTP-based transactional email), which uses HTTPS port 443 — always available. Replaced the SMTP implementation with a clean `httpx.AsyncClient.post()` call to `https://api.resend.com/emails`.

Challenge 3: Google SSO — Password-less Account Security

Problem: Students who registered via Google OAuth had no `password_hash`. If they later tried the "Forgot Password" flow, there was nothing to reset.
Solution: On Google OAuth registration, a **cryptographically random temporary password** is generated using `secrets.token_urlsafe(32)` and hashed before being stored. This allows the standard password reset flow to function normally if the user later wants to set a real password.

Challenge 4: Behavioral Proctoring — Submission ID Timing

Problem: The `BehaviorLog` needed to reference a `submission_id`, but the submission record is only created when the student **starts** the exam — the behavior logging begins before that.
Solution: Changed the lookup strategy. During the exam, behavior logs are stored and retrieved by `(user_id, exam_id)` pair. The `submission_id` is bound **late** (updated in the log when the student formally starts). This eliminates the circular dependency.

Challenge 5: Import Resolution After Refactoring

Problem: After physically moving 40+ frontend files across directories (hooks, services, components), all existing `import` paths broke, causing the Vite build to fail across nearly every component.
Solution: Used a targeted `sed` command to atomically rewrite all import paths across every affected file in one pass. The `npm run build` was used as the ground-truth validator after each batch of path corrections.

---

13. Intern Responsibilities

As an intern on the ExamGuard AI project, the following responsibilities were assigned and fulfilled:

Full-Stack Development

- Designed and implemented the complete full-stack system architecture
- Built **17 Admin portal pages** and **12 Student portal pages** in React
- Implemented **40+ FastAPI REST API endpoints** across **16 route files**
- Created **5 MongoDB document models** with strict Beanie ODM schemas

Core Feature Engineering

- Engineered the **Behavioral Proctoring Engine** — delta-based telemetry system with real-time anomaly scoring
- Built the **AI Test Generator** — curated question bank with 200+ templates across 5+ technical topics
- Integrated the **Judge0 code execution** proxy endpoint for in-browser code IDE
- Implemented **Google OAuth2 SSO** with server-side token exchange
- Built the **Resend API email service** for transactional password reset emails
- Developed the **Cold-Start Resilience** system for Render free-tier hibernation

Authentication & Security

- Implemented dual-isolated JWT authentication schemes (Student + Admin)
- Applied `bcrypt` password hashing, timezone-safe datetime comparisons, and strict Pydantic input validation
- Configured CORS, environment variable management, and secret key isolation

DevOps & Deployment

- Dockerized the FastAPI backend and deployed to **Render**

- Configured **Vercel** frontend deployment with environment variable injection
- Set up **MongoDB Atlas** free cluster with network access rules
- Established end-to-end CI/CD via GitHub integration

Architectural Refactoring (10-Phase Audit)

- Led a comprehensive 10-phase codebase audit
- Restructured both frontend and backend to strict domain-driven architecture
- Authored `DEVELOPER_GUIDE.md`, `API_REFERENCE.md`, and `DEPLOYMENT_GUIDE.md`

---

14. Key Learnings

Domain	Learning
FastAPI	Async route handlers, Beanie ODM, Pydantic V2 schema design, JWT middleware
React Architecture	Domain-driven component design, Custom hooks as business logic containers, Context API for auth state
MongoDB	Document schema design for deeply nested data, compound index optimization, Beanie ODM patterns
Security	JWT isolation between roles, bcrypt hashing, OAuth2 PKCE flow, timezone-safe datetime comparisons
DevOps	Docker containerization, Vercel/Render CI/CD pipelines, Atlas network configuration
API Design	REST convention adherence, consistent error response structure, HTTP status code semantics
Problem Solving	Debugging production SMTP blocks, resolving proctoring data race conditions, fixing import paths at scale
Engineering	How to systematically audit, refactor, and document a large codebase

---

15. Future Scope

Feature	Priority	Description
ML Anomaly Detection	High	Train a scikit-learn classifier on historical behavioral logs to replace the rule-based anomaly scoring
Video Proctoring	High	Integrate WebRTC + MediaRecorder API for optional webcam

Feature	Priority	Description
		snapshots during exam sessions
Plagiarism Engine	Medium	Run MOSS (Measure of Software Similarity) on submitted code to detect copy-paste plagiarism
Email Notifications	Medium	Send automated score reports and exam reminders to students
Mobile App	Medium	Build a React Native app for students to access the platform on mobile
Question Bank Editor	Low	Admin UI to add/edit questions directly into the `template_repository.py` question bank
WebSocket Live Monitoring	Low	Replace HTTP polling with WebSocket for truly live proctoring feeds
Multi-Tenant Support	Low	Allow multiple institutions to run isolated instances on the same backend

---

16. Conclusion

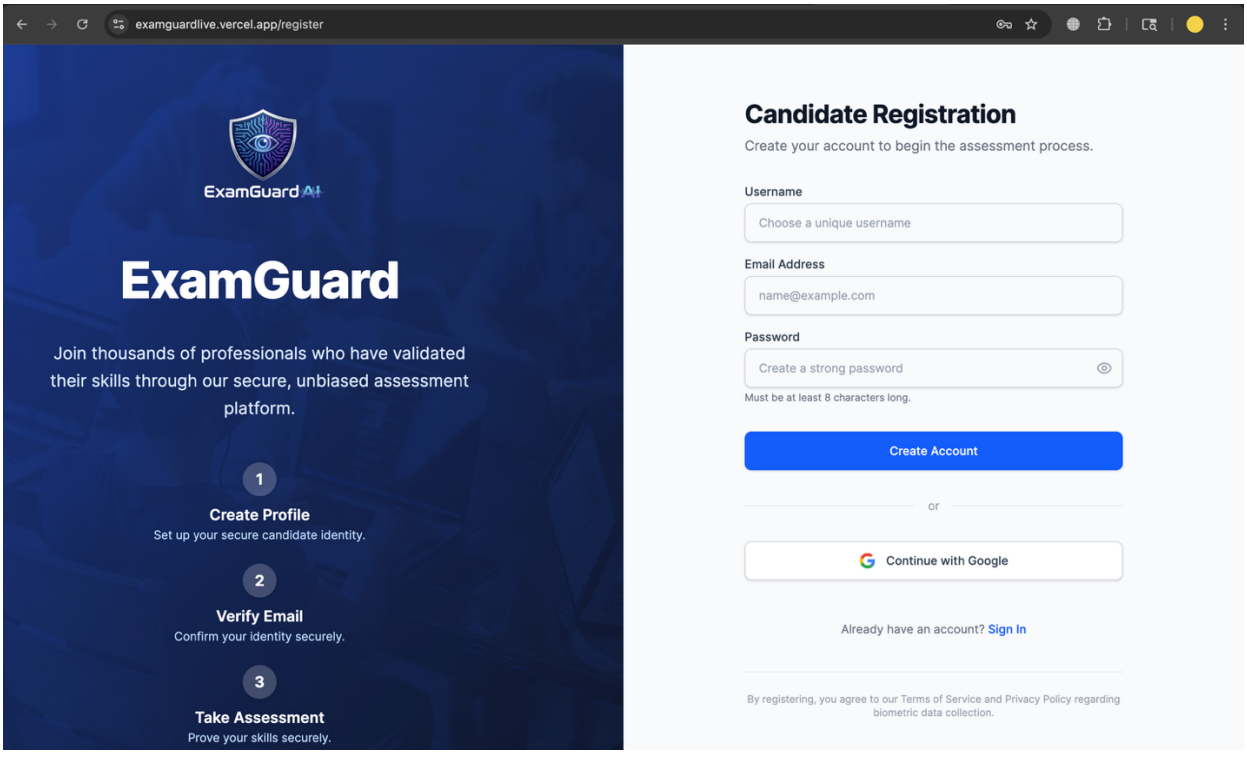
ExamGuard AI successfully demonstrates the development of a production-grade, enterprise-scale full-stack web application from concept to deployment. The project integrates a diverse set of modern technologies — React 19, FastAPI, MongoDB, Docker, Google OAuth, Judge0, and Resend — into a cohesive, well-architected system.

The internship provided deep, practical experience in:

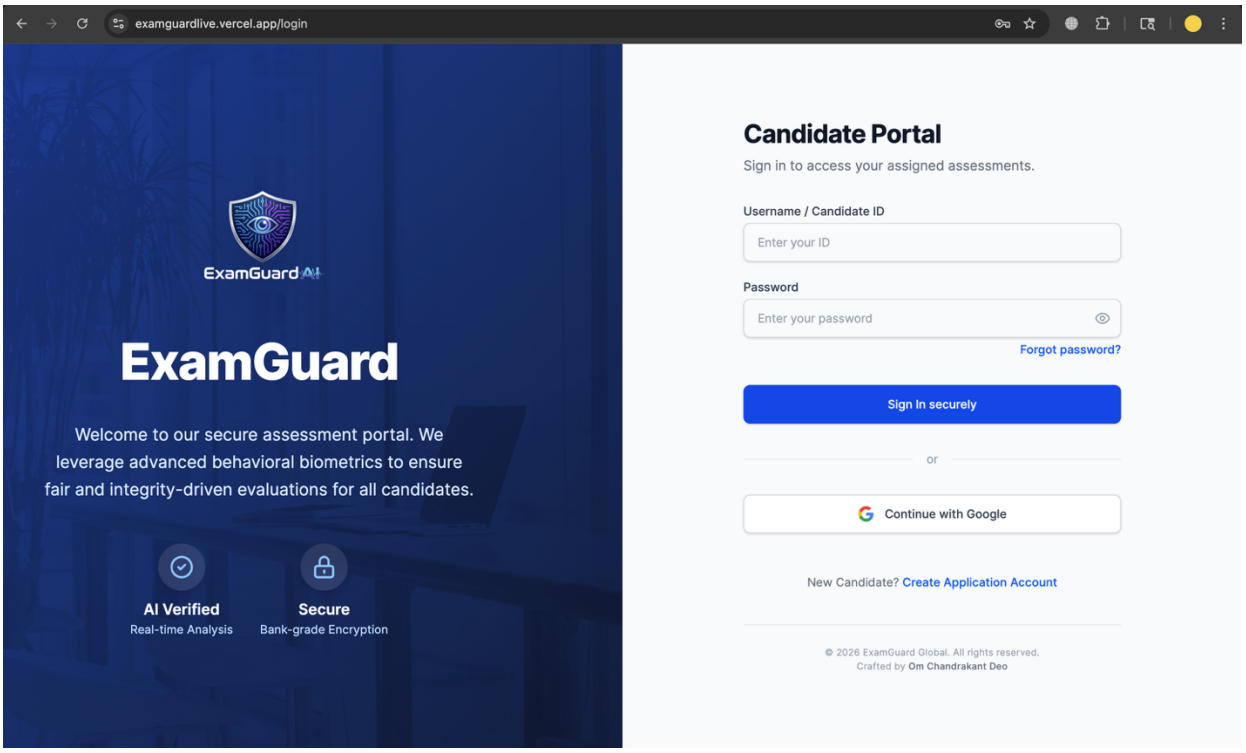
- Designing clean REST APIs that are secure, consistent, and well-documented
- Managing complex state in large React applications through custom hooks and context providers
- Operating cloud infrastructure (Vercel + Render + MongoDB Atlas)
- Performing systematic architectural audits and large-scale codebase refactoring
- Debugging production issues including SMTP blocks, timezone crashes, and data race conditions

The resulting platform is fully functional, deployed to production, and maintained under version control with professional documentation that enables any new developer to understand, extend, and deploy the system independently.

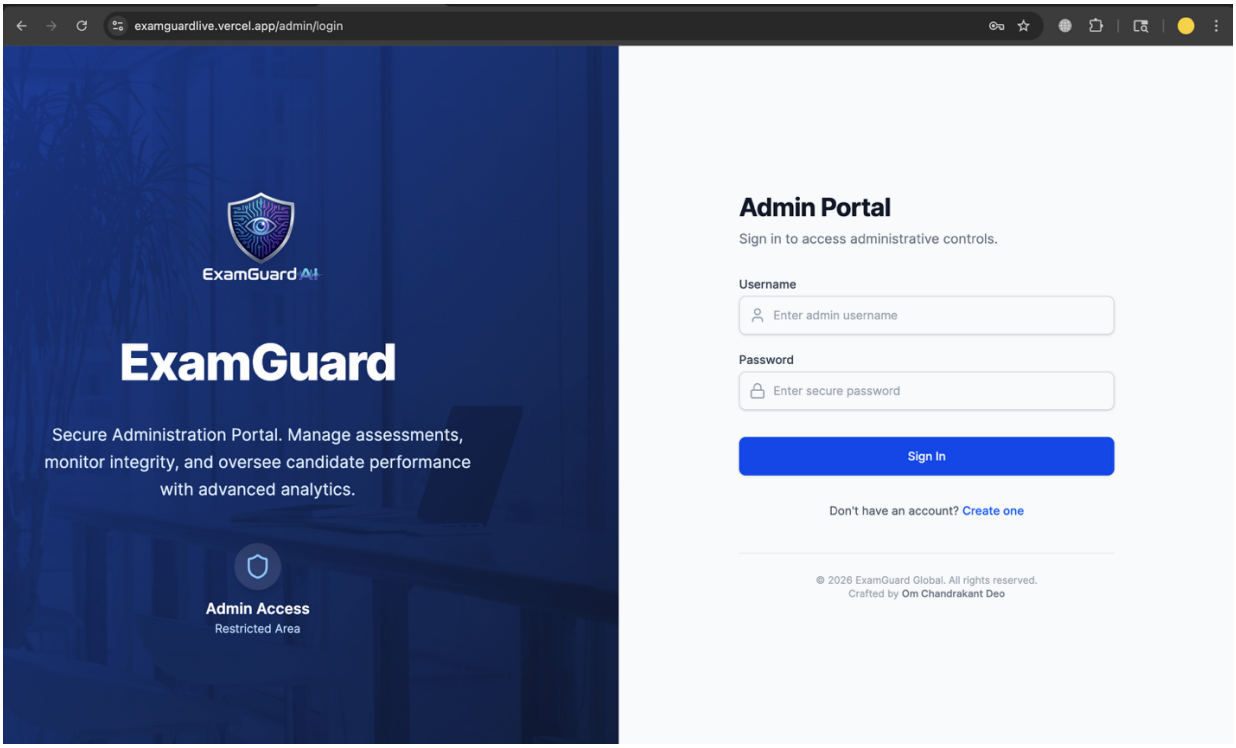
Candidate Registration:



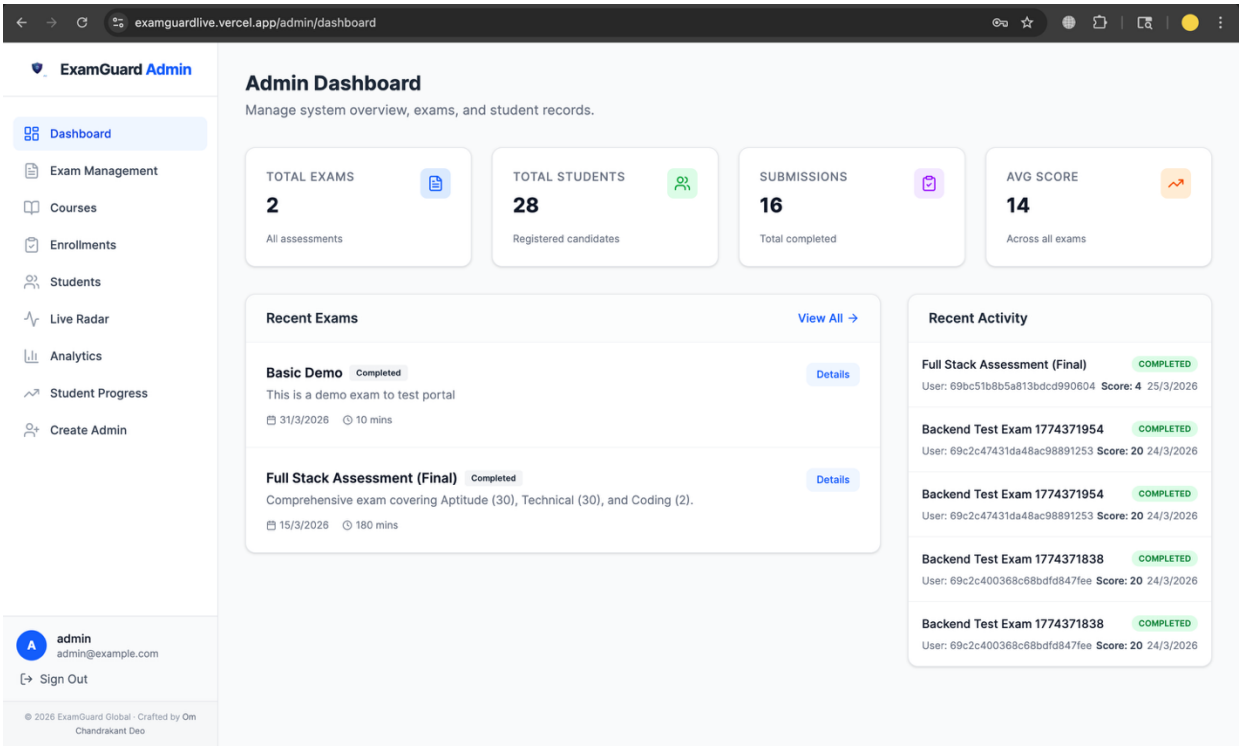
Candidate Login:



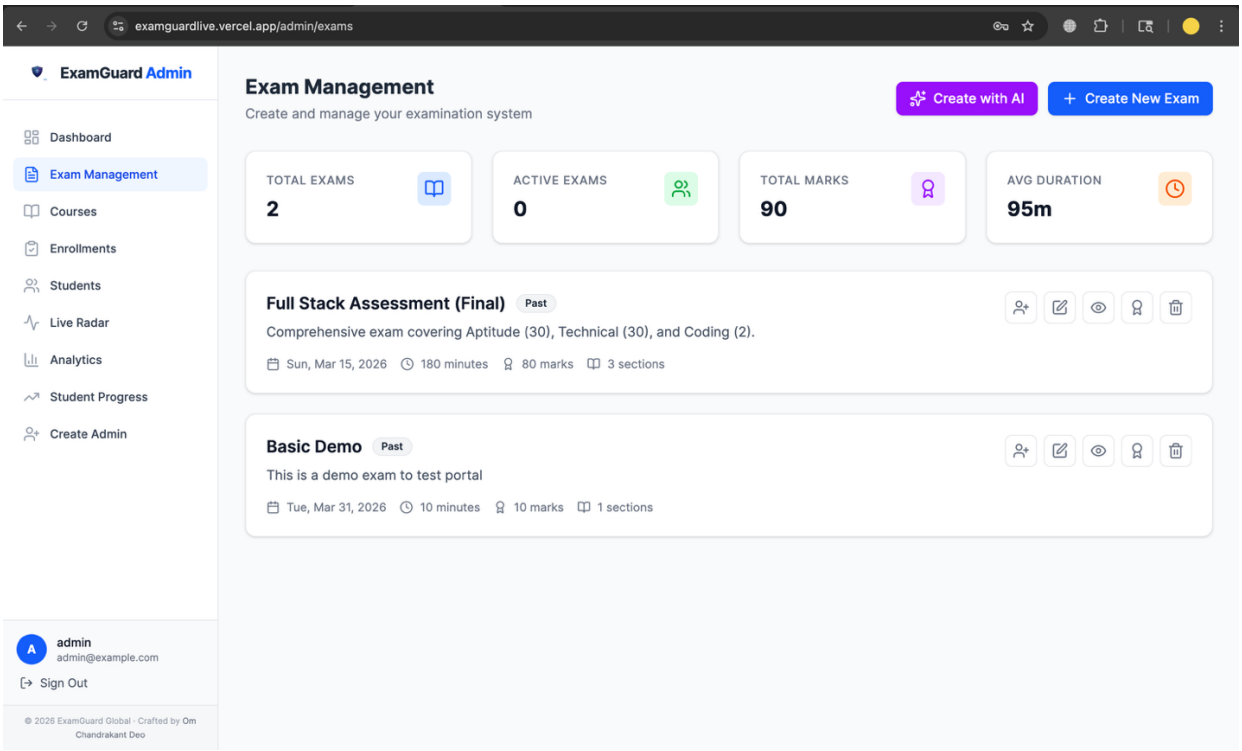
Admin Login:



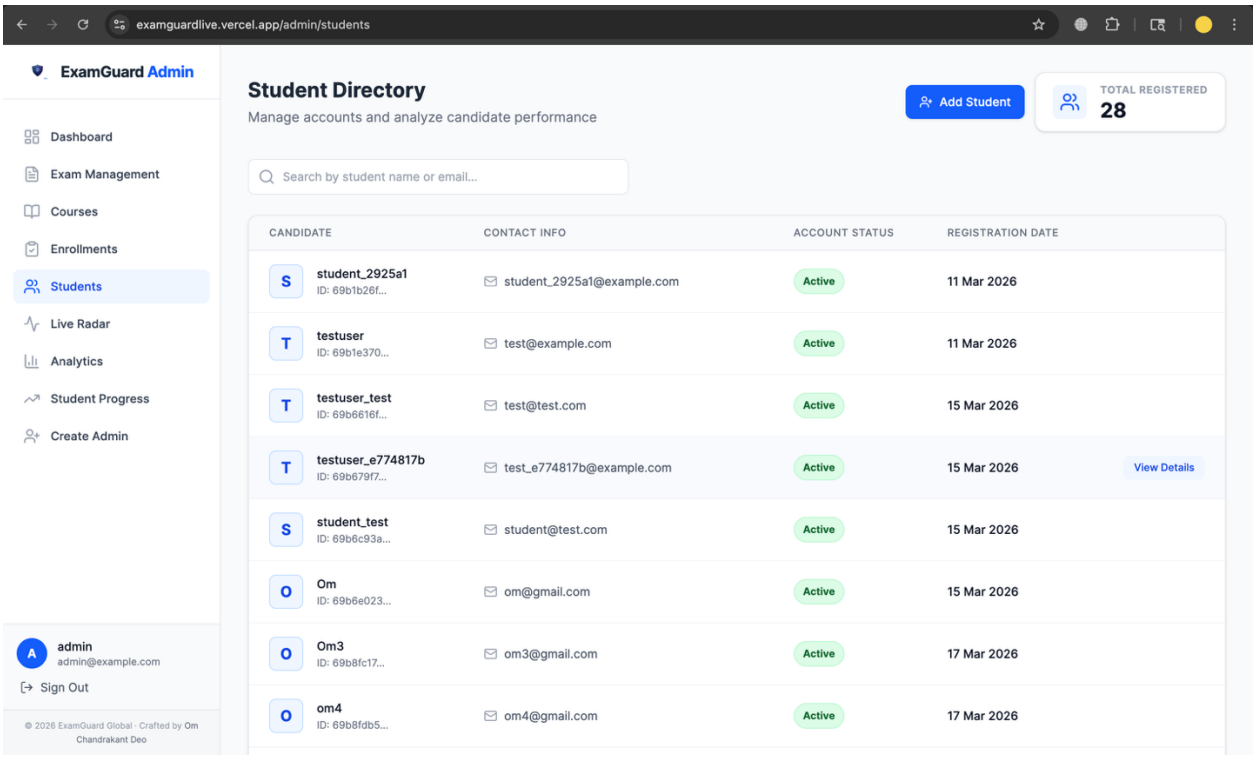
Admin Dashboard:



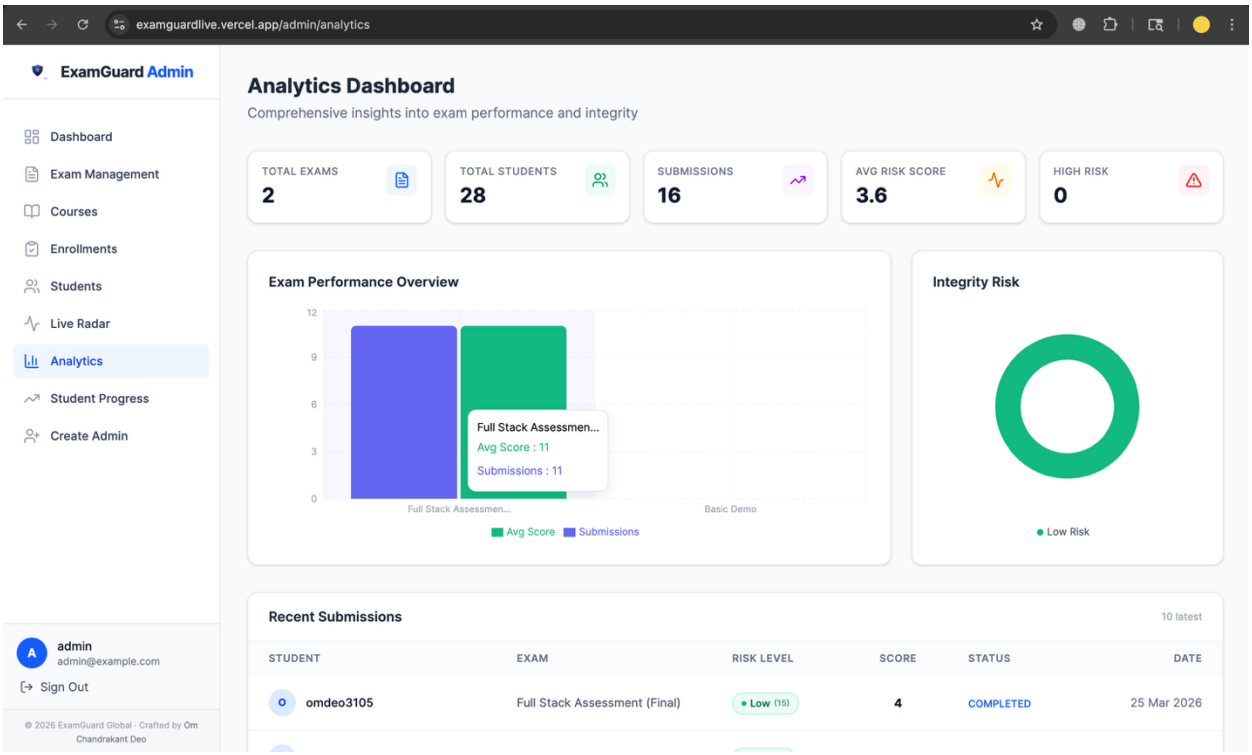
Exam Management:



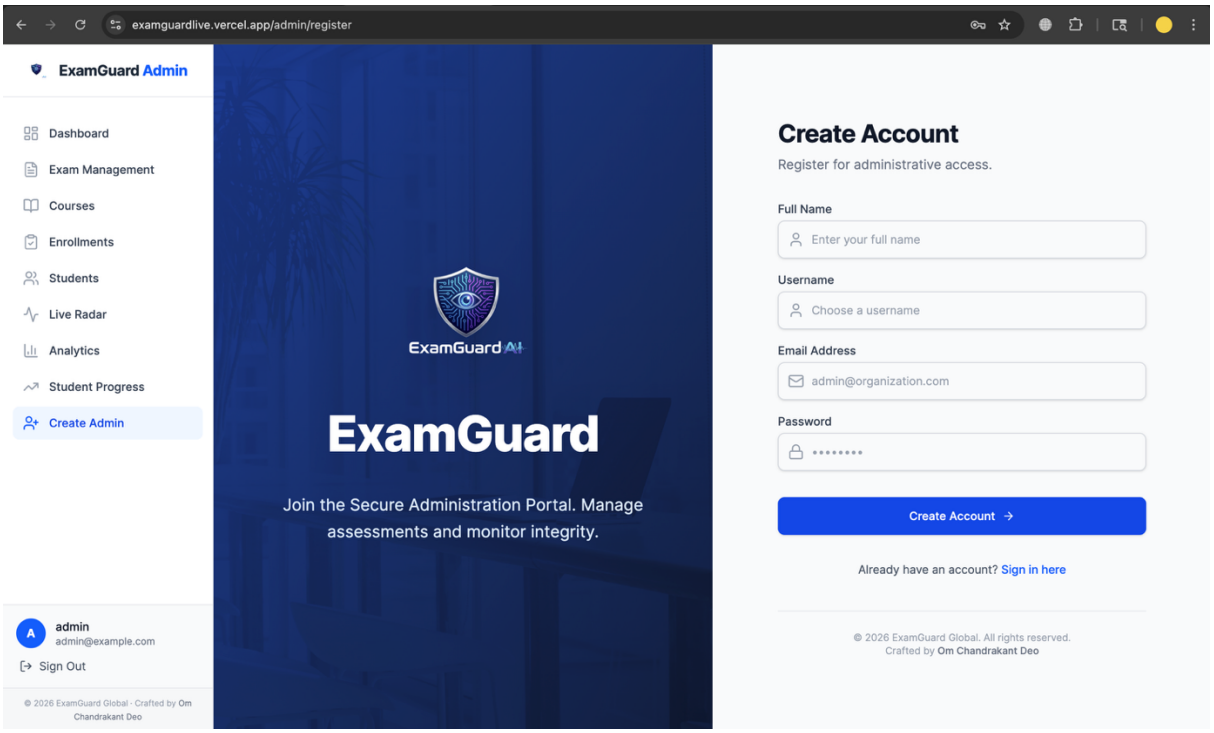
Student Management:



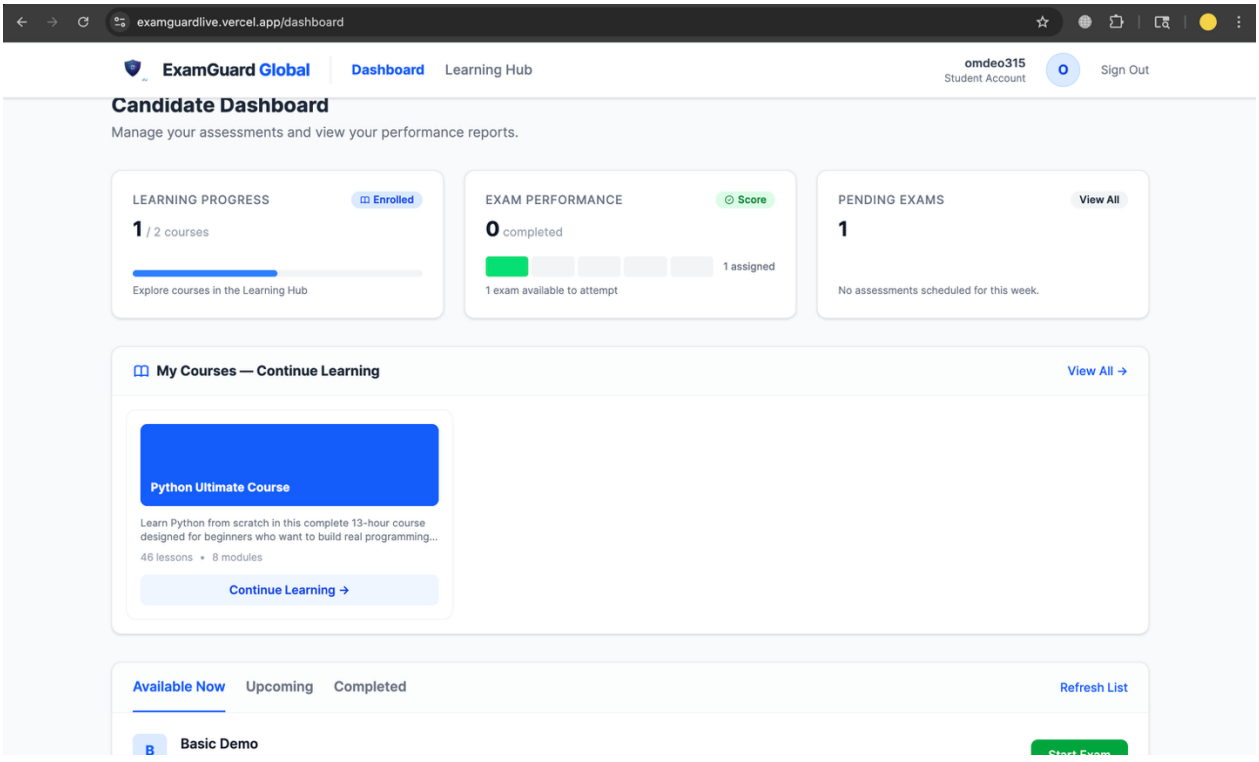
Analytics Dashboard:

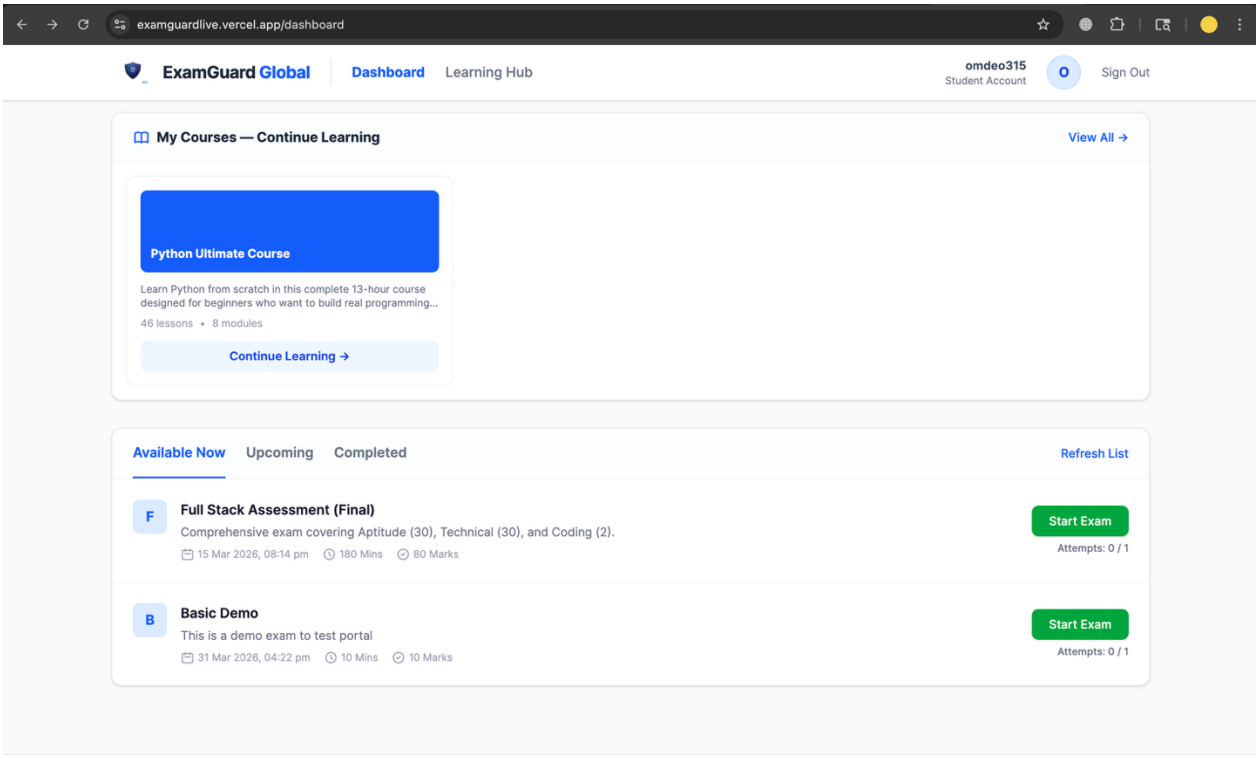


Create New Admin:

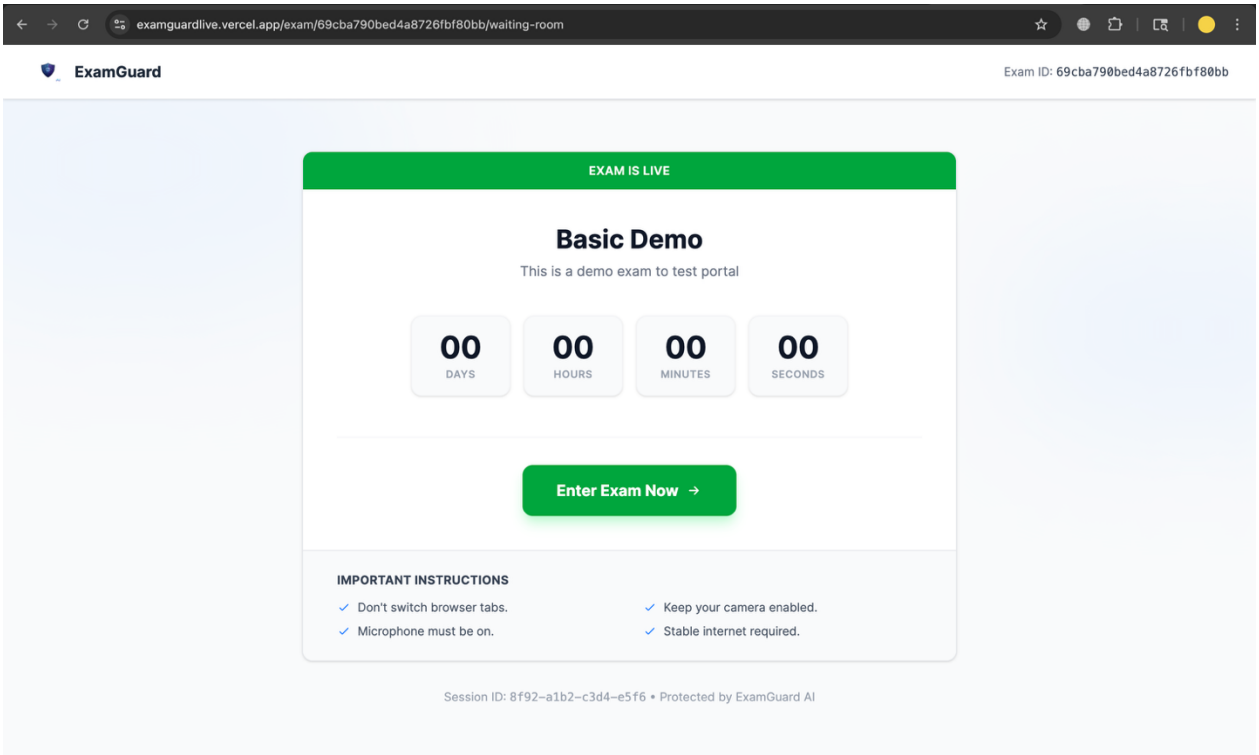


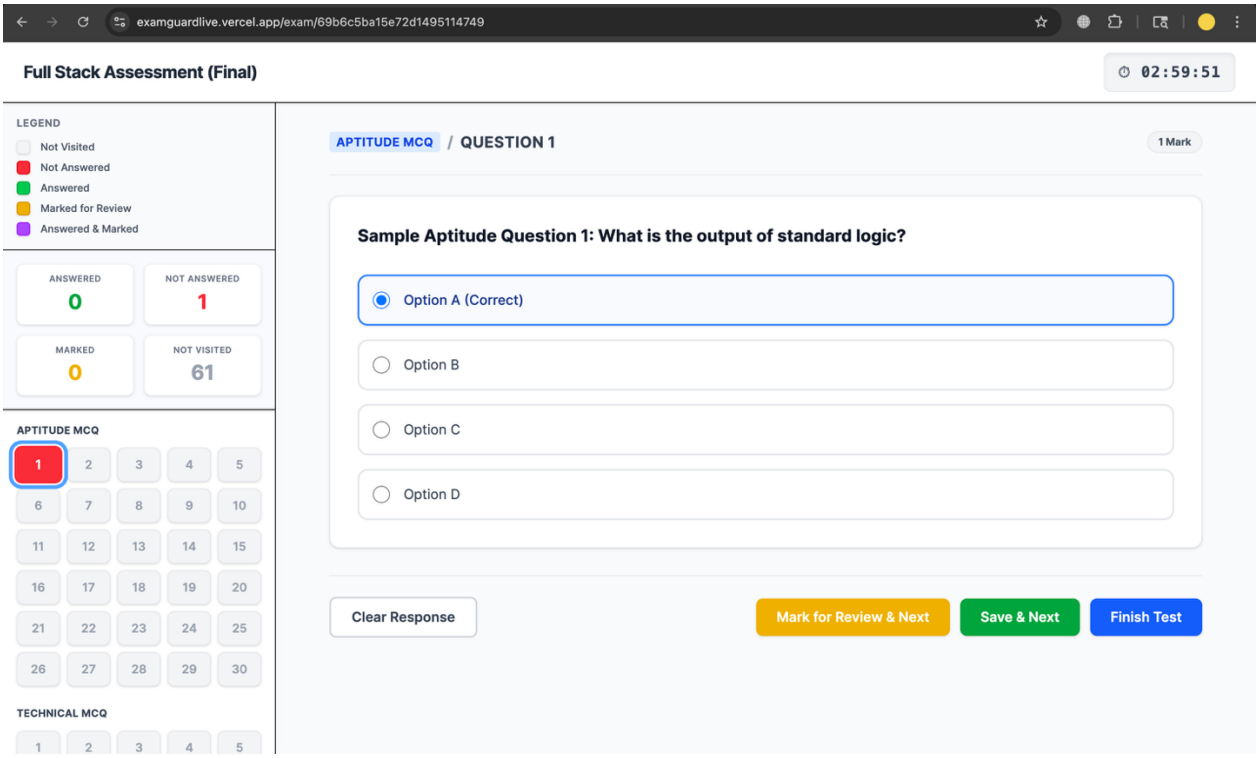
Candidate Dashboard:

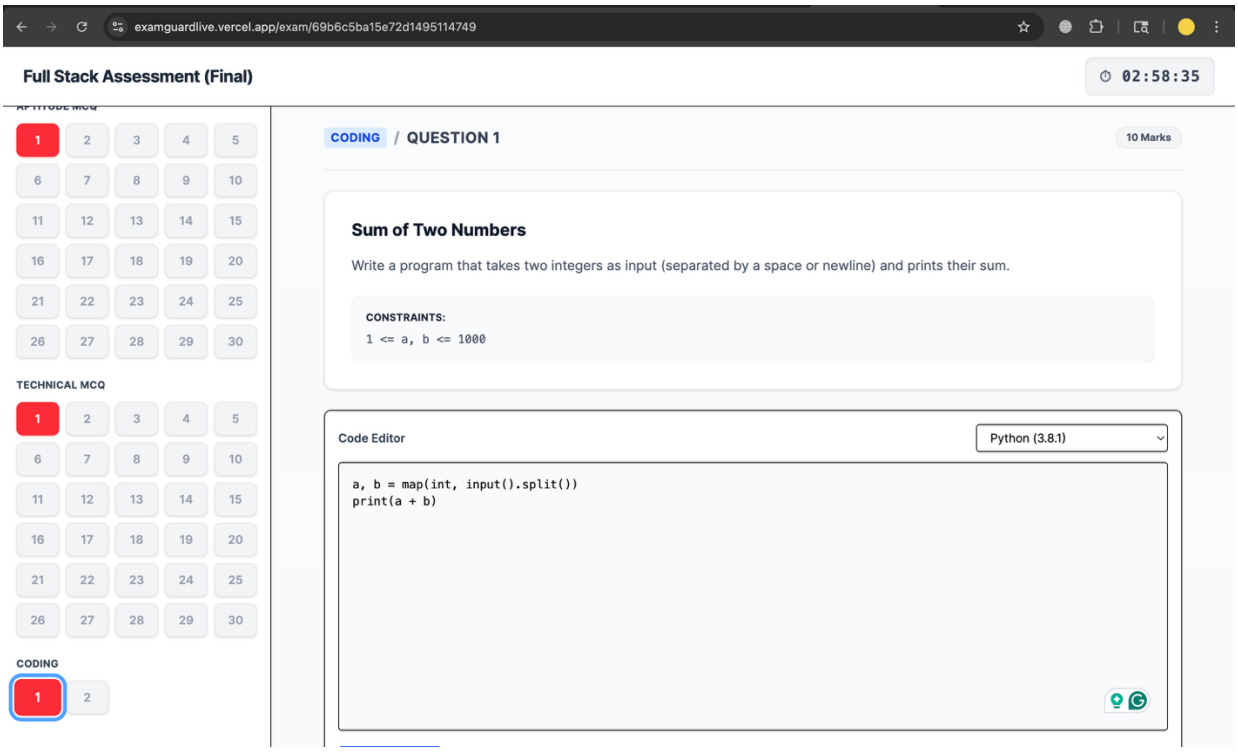


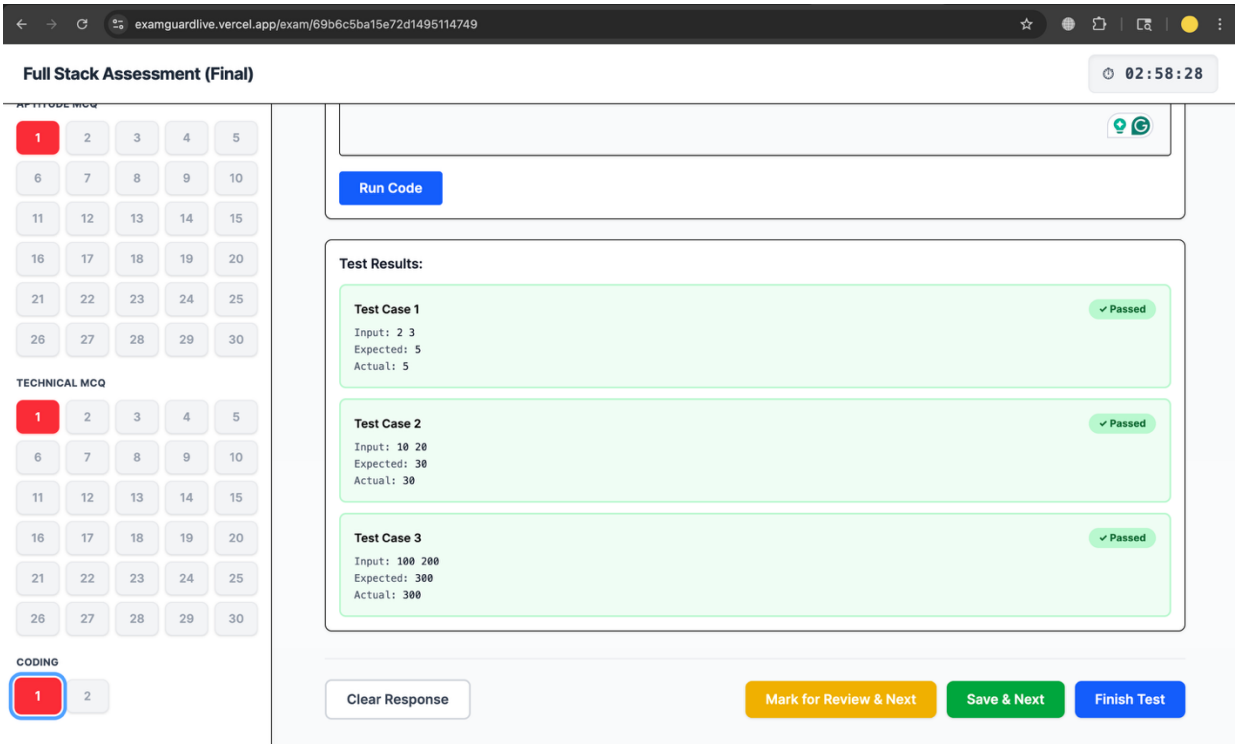

examguardlive.vercel.app/dashboard
ExamGuard Global
Dashboard
Learning Hub
omdeo315
Student Account
O
Sign Out
My Courses — Continue Learning
View All →
Python Ultimate Course
Learn Python from scratch in this complete 13-hour course designed for beginners who want to build real programming...
46 lessons • 8 modules
Continue Learning →
Available Now
Upcoming
Completed
Refresh List
F
Full Stack Assessment (Final)
Comprehensive exam covering Aptitude (30), Technical (30), and Coding (2).
15 Mar 2026, 08:14 pm
180 Mins
80 Marks
Start Exam
Attempts: 0 / 1
B
Basic Demo
This is a demo exam to test portal
31 Mar 2026, 04:22 pm
10 Mins
10 Marks
Start Exam
Attempts: 0 / 1

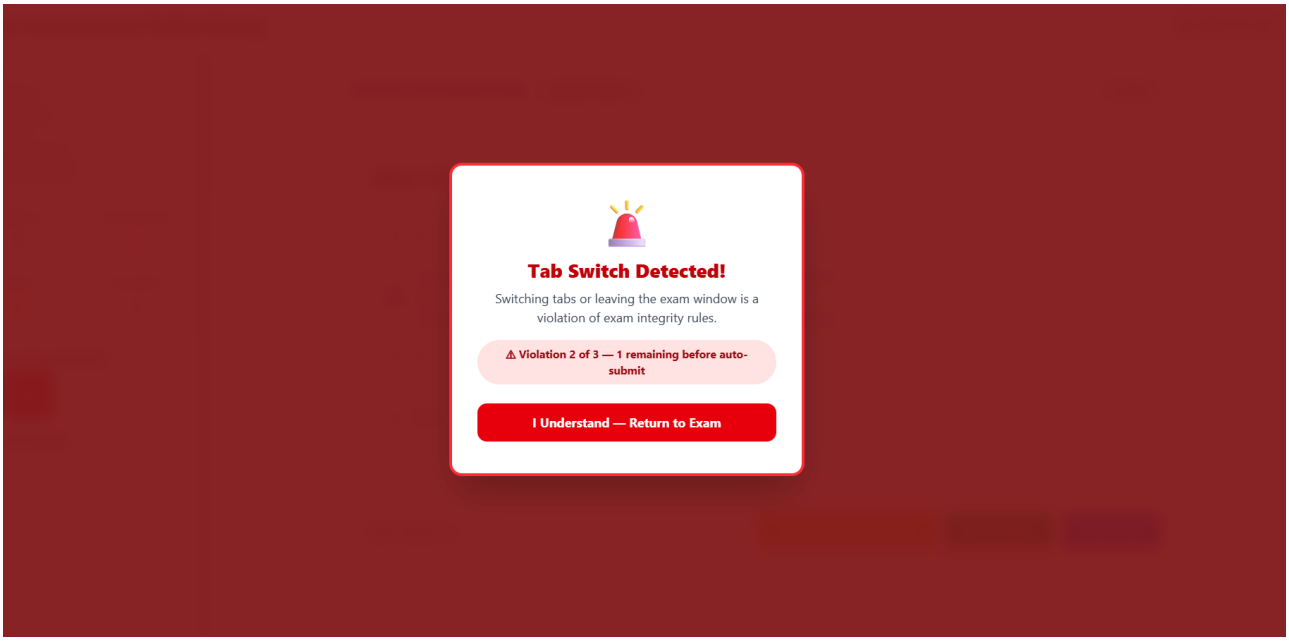
Exam Flow:

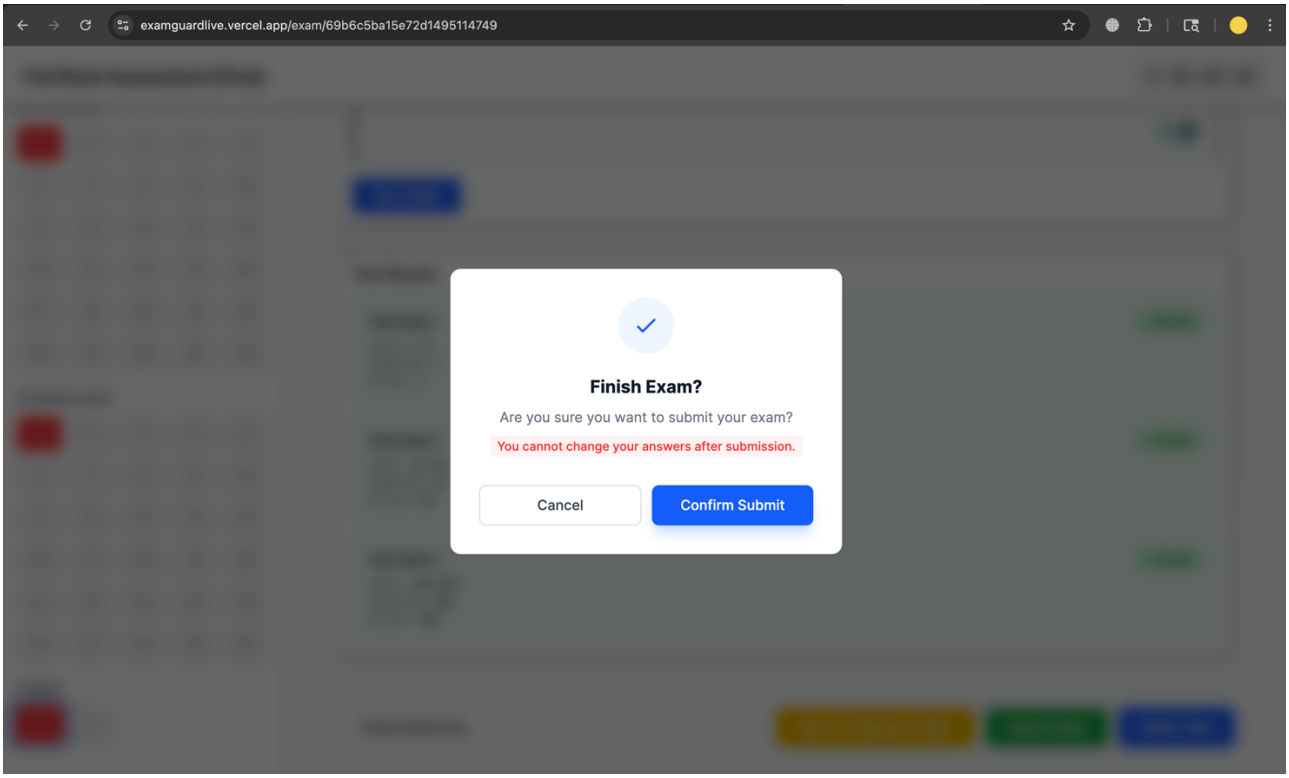

examguardlive.vercel.app/exam/69cba790bed4a8726fbf80bb/waiting-room
ExamGuard
Exam ID: 69cba790bed4a8726fbf80bb
EXAM IS LIVE
Basic Demo
This is a demo exam to test portal
00
DAYS
00
HOURS
00
MINUTES
00
SECONDS
Enter Exam Now →
IMPORTANT INSTRUCTIONS
Don't switch browser tabs.
Keep your camera enabled.
Microphone must be on.
Stable internet required.
Session ID: 8f92-a1b2-c3d4-e5f6 • Protected by ExamGuard AI

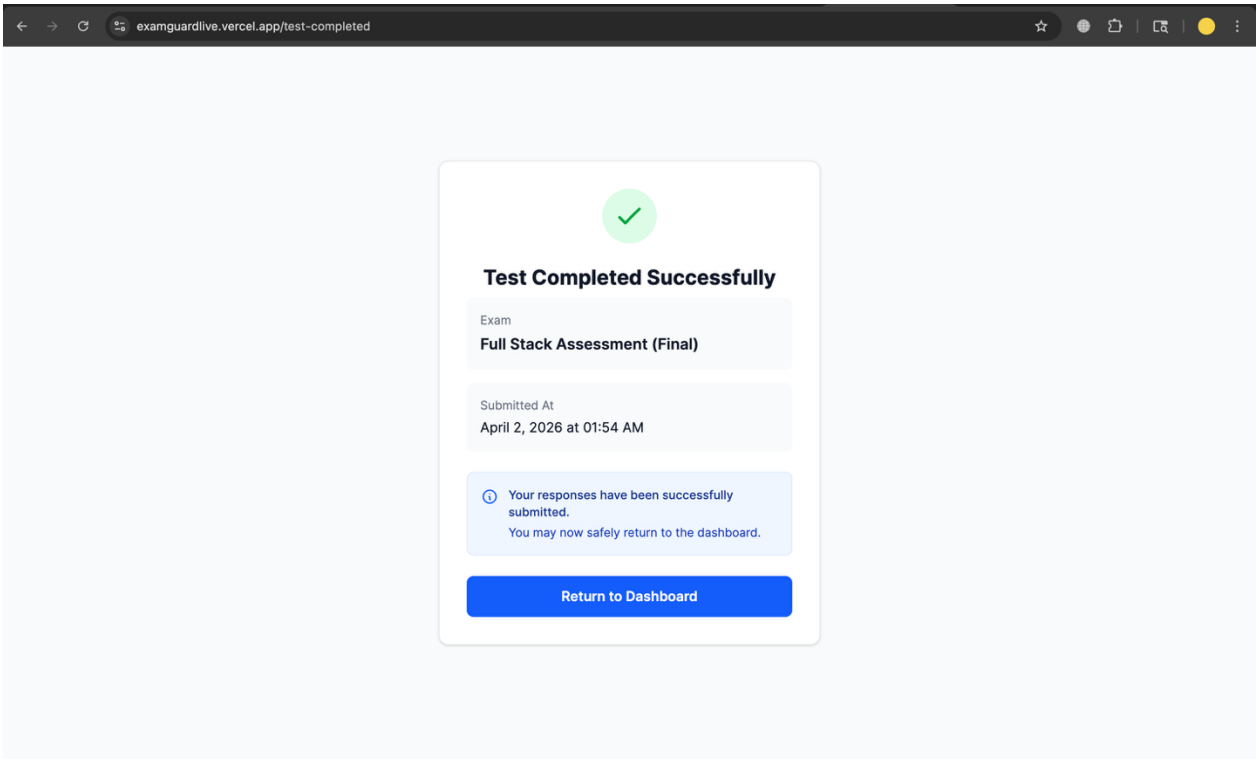

examguardlive.vercel.app/exam/69b6c5ba15e72d1495114749
Full Stack Assessment (Final)
02:59:51
LEGEND
Not Visited
Not Answered
Answered
Marked for Review
Answered & Marked
ANSWERED 0
NOT ANSWERED 1
MARKED 0
NOT VISITED 61
APTITUDE MCQ
TECHNICAL MCQ
APTITUDE MCQ / QUESTION 1
1 Mark
Sample Aptitude Question 1: What is the output of standard logic?
Option A (Correct)
Option B
Option C
Option D
Clear Response
Mark for Review & Next
Save & Next
Finish Test


examguardlive.vercel.app/exam/69b6c5ba15e72d1495114749
Full Stack Assessment (Final)
02:58:35
TECHNICAL MCQ
CODING
CODING / QUESTION 1
10 Marks
Sum of Two Numbers
Write a program that takes two integers as input (separated by a space or newline) and prints their sum.
CONSTRAINTS:
1 <= a, b <= 1000
Code Editor
Python (3.8.1)
a, b = map(int, input().split())
print(a + b)


examguardlive.vercel.app/exam/69b6c5ba15e72d1495114749
Full Stack Assessment (Final)
02:58:28
TECHNICAL MCQ
CODING
Run Code
Test Results:
Test Case 1
Passed
Input: 2 3
Expected: 5
Actual: 5
Test Case 2
Passed
Input: 10 20
Expected: 30
Actual: 30
Test Case 3
Passed
Input: 100 200
Expected: 300
Actual: 300
Clear Response
Mark for Review & Next
Save & Next
Finish Test


Tab Switch Detected!
Switching tabs or leaving the exam window is a violation of exam integrity rules.
⚠ Violation 2 of 3 — 1 remaining before auto-submit
I Understand — Return to Exam


examguardlive.vercel.app/exam/69b6c5ba15e72d1495114749
Finish Exam?
Are you sure you want to submit your exam?
You cannot change your answers after submission.
Cancel
Confirm Submit


examguardlive.vercel.app/test-completed
Test Completed Successfully
Exam
Full Stack Assessment (Final)
Submitted At
April 2, 2026 at 01:54 AM
Your responses have been successfully submitted.
You may now safely return to the dashboard.
Return to Dashboard

Outcomes

The following outcomes were successfully achieved by the end of the internship:

- A **fully functional dual-portal web application** (Admin and Student) was developed, deployed, and made accessible through public URLs, demonstrating end-to-end system integration.
- Implementation of a complete **examination workflow** supporting both MCQ and coding formats, including automated evaluation, result generation, and submission tracking.
- Development of a **real-time proctoring system** capable of detecting tab-switching, clipboard actions, and behavioral anomalies, with all violations logged and available for administrative review.
- A comprehensive **Learning Management System (LMS)** enabling course creation, content delivery (videos, notes, quizzes), student enrollment, progress tracking, and certificate generation.
- Creation of **interactive analytics dashboards** for administrators, providing insights into student performance, exam statistics, and integrity risk scores with efficient and responsive data visualization.
- Successful integration of external services such as **Judge0 for secure code execution**, ensuring accurate and scalable evaluation of coding assessments.

- Deployment of the system using **cloud platforms (Render and Vercel)**, ensuring accessibility, reliability, and real-world usability of the application.
- Significant personal development in **full-stack development, asynchronous programming, API integration, system design, and cloud deployment practices**, along with improved problem-solving and debugging skills.

Conclusion

The internship on **ExamGuard AI** proved to be a highly enriching and transformative experience, effectively bridging the gap between academic learning and real-world industry practices. The project involved solving complex challenges such as designing secure multi-role authentication systems, implementing scalable backend services, integrating third-party APIs, and deploying a full-stack application to cloud platforms—all of which were successfully accomplished.

Working in a collaborative and Agile-driven environment enhanced both technical proficiency and teamwork skills. The hands-on experience with modern technologies such as **FastAPI, React 18, MongoDB, and cloud deployment platforms** provided a strong foundation in full-stack development and system design.

Furthermore, the project highlighted the importance of building **secure, scalable, and user-centric applications** in the EdTech domain. ExamGuard AI stands as a demonstration of how the right combination of architecture, technology stack, and problem-solving approach can result in a robust platform capable of meeting the evolving demands of digital education.

Overall, this internship not only strengthened my technical capabilities but also improved my ability to approach problems systematically, making it a significant step toward a professional career in software engineering.

Appendix C

Daily Work Sheet

Duration: (in Hours)

Day No.	Activity Planned	Activity Completed Status	Student Signature	Mentor Signature
Day 1				
Day 2				
Day 3				
Day 4				
Day 5				
Day 6				
Day 7				
Day 8				
Day 9				
Day 10				
Day 11				
Day 12				
Day 13				
Day 14				
Day 15				
Day 16				
Day 17				
Day 18				
Day 19				
Day 20				
Day 21				
Day 22				
Day 23				
Day 24				

INTERNSHIP – DAILY WORKING REPORT SHEET

- **Lab Session Details**

Date	Day	Time (From – To)

- **Project Information**

Title of Project:	ExamGuard AI — AI-Powered Online Examination Management Platform
Objective Name	
Title of Work Performing:	

- **Project Parameters Involved**
  ☐ Frontend ☐ Backend ☐ Database ☐ Machine Learning / AI ☐ API Integration
  ☐ Simulation / Hardware ☐ Testing / Optimization

- **Student Information**

Roll Number	Student Name	Signature

- **Work Done During Lab Session / Work Completion**

(Describe the exact work performed during this lab duration)

- **Remarks by Verifying Faculty (Satisfactory / Not-Satisfactory)**

Appendix D

Assignments during internship

Sr. No.	Name of Assignment/Activity	Status (Completed/NC)	Remark